

5. Shukla, A., “Practical Fracture Mechanics in Design”, 2nd Ed., CRC Press. 1989.
6. Bazant, Z. P. and Cedoliin, L., “Stability of Structures: Elastic, Inelastic, Fracture and Damage Theories”, World Scientific Publishers. 2010.

Evaluation Scheme

S.N o.	Evaluation Elements	Weightage (%)
1.	MST	30
2.	EST	45
3.	Sessionals (Assignments/Projects/Tutorials/Quizes)	25

UME746 : System Modelling and Simulation

L	T	P	Cr
2	0	2	3.0

Course Objective: The objective for this course is to develop an understanding of the cause-and-effect interaction of different components of a system. This understanding will include concepts such as analysis of rigid bodies, structural systems, hydraulic systems, thermal systems, electronic and mechatronic systems, multibody systems, thermo-mechanical systems and control strategies.

Syllabus

Modelling in Multi-Energy Domain through Bond Graphs: Introduction to bond graphs, power variables of bond graphs and models of simple circuits, reference power directions, bond graph elements and their constitutive relations, causality, generation of system equations from bond graph models, the idea of activation.

System Modelling: Modelling of a system of rigid bodies, structural systems, hydraulic systems, thermal systems, electronic and mechatronic systems.

Modelling of Multibody Systems: Mechanisms, manipulators and vehicles.

Advanced topics in Bond Graph Modelling of Physical Systems: Elements of multi-bond graphs, thermo-mechanical bond graphs and continuous systems and other systems of typical interest.

Control System: Modelling systems for control strategies in physical domain i.e., P, PI, PID, overwhelming control. Stability of systems from signal flow graph using Routh's criterion.

Laboratory Work: Bond graph modelling of rigid bodies, structural systems, hydraulic systems, thermal systems, electronic and mechatronic systems using MATLAB Simulink.

Research Assignment

Numerical prototyping as modelling for design and synthesis using computational tools for the systems like bicycle vehicle, parallel manipulator with overwhelming control, Rapson slide, inverted pendulum, car moving over bump etc.

Course Learning Objectives (CLO)

The students will be able to:

1. frame bond graph models of systems using power variables, reference power directions, causality.
2. generate the system equations from bond graph models.
3. develop bond graph models of structural/thermal systems
4. create different control systems and stability analysis using bond graph.

Text Books

1. Mukherjee, R. Karmakar, A.K. Samantaray, Bond Graph in Modeling, Simulation and fault Identification, CRC Press, FL (2006).
2. D.C. Karnopp, D.L. Margolis, R.C. Rosenberg, System Dynamics, Modeling and Simulation of Mechatronic Systems, John Wiley & Sons, NY (2000).

Reference Books

1. B OuldBouamama, J Thoma, Modelling and Simulation in Thermal and Chemical Engineering: A Bond Graph Approach, Springer, New York (2000).
2. Dean Karnopp, Vehicle Dynamics, Stability, and Control, CRC Press, (2013).
3. R. Merzouki, A.K. Samantaray, P.M. Pathak, B. OuldBouamama, Intelligent Mechatronic Systems: Modeling, Control and Diagnosis, Springer, New York (2012).
4. Borutzky, W., Bond graphs: a methodology for modelling multidisciplinary dynamic systems, SCS Publishing House, Erlangen, San Diego (2004).

Evaluation Scheme:

Sl. No.	Evaluation Elements	Weightage (%)
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